

Beyond Kernel Rank: Optimal-Fiber Geometry and Rank-Two Topology of Minimum-Completion Quotients

Alp Eren Bütün
Independent Researcher, Türkiye

August 2026

Abstract

Let $A \in \mathbb{Z}^{k \times d}$ and let r be a completable residual. The minimum-completion problem

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d, Ax = -r\}$$

has an optimal Parikh fiber

$$\mathcal{O}_A(r) = \{x \in \mathbb{N}^d : Ax = -r, \mathbf{1}^\top x = \ell_A(r)\}.$$

The associated shortest-prefix down-set carries a natural cubical structure, and equal-length kernel trades produce a global quotient $I_A(r)$ recording all shortest completion schedules modulo residual-preserving identifications. This paper studies the topology of that quotient beyond the rank-one regime.

The first result is an optimal-fiber compression theorem. If

$$L_A(r) = \langle \mathcal{O}_A(r) - \mathcal{O}_A(r) \rangle_{\mathbb{Z}}$$

is the active trade lattice and $P_A(r) = \mathcal{O}_A(r) - x_0$ is a normalized optimal fiber, then

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r),$$

where $\mathcal{N}(P)$ is the periodic nerve of the translates of P . Thus the global homotopy type is determined by the translation geometry of the finite optimal fiber alone. The cellular chains of this compressed quotient are then identified with a principal, dehomogenized slice of the Bayer–Sturmfels Taylor resolution of the lattice ideal of $L_A(r)$; a character-twisted version controls local-system homology.

For active rank two we obtain a complete homotopy classification. The first Betti number is always 2 or 3. If the normalized fiber contains no nondegenerate lattice parallelogram, then

$$I_A(r) \simeq \bigvee^{\beta_1(I_A(r))} S^1.$$

If it contains a nondegenerate parallelogram, then

$$I_A(r) \simeq T^2 \vee \bigvee^{\beta_2(I_A(r))-1} S^2.$$

Consequently the possible active-rank-two homotopy types are exactly

$$S^1 \vee S^1, \quad S^1 \vee S^1 \vee S^1, \quad T^2 \vee \bigvee^m S^2 \quad (m \geq 0).$$

All of them are realized by explicit completion-total scalar systems. In particular, the second Betti number is unbounded while the active rank and, in the toral branch, the fundamental group remain fixed. A separate fixed-system family shows that even the matrix, the full equal-length kernel, the active lattice, and the minimum completion length can remain unchanged while the global topology changes from $S^1 \vee S^1$ to T^2 as the residual varies.

Keywords. deterministic pushdown normalization; shortest completion; optimal Parikh fiber; lattice ideals; cellular resolutions; nerve complexes; homotopy classification; Freiman dimension; lattice polygons.

1 Introduction

The topology studied here arose from a completion problem rather than from an abstract topological construction. The original source was a completion-aware 2:1 deterministic pushdown automaton: after a prefix is read, its control state and stack encode the symbol demand still required to restore weighted balance. Passing from the binary balance law to a matrix counting system replaces a unique shortest completion vector by a finite family of equally short vectors. That replacement is the beginning of higher-dimensional completion geometry.

For an integer matrix $A \in \mathbb{Z}^{k \times d}$, the basic objects are the minimum length $\ell_A(r)$, the optimal fiber $\mathcal{O}_A(r)$, the equal-length kernel

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix},$$

and the filled shortest-prefix quotient $I_A(r)$. In rank zero the geometry is a literal box. In rank one the nontrivial quotient is rigid: it is a circle. The first genuinely higher-rank question is therefore whether the rank of K_A , or of the portion of it visible at a given residual, determines the topology of $I_A(r)$.

It does not. The main conclusion of the paper is that rank controls the onset of topology, but the *translation geometry of the optimal completion fiber* controls its type. In active rank two the distinction can be read from a simple planar feature: the presence or absence of a nondegenerate lattice parallelogram.

Main contributions. The paper is organized around the following results.

- (i) **Optimal-fiber compression.** All kernel collisions that actually occur inside the shortest-prefix region are generated by differences of optimal endpoints. The global cubical quotient therefore admits a finite-fiber nerve model

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r).$$

- (ii) **Active lattice and lattice-convexity.** The normalized fiber is lattice-convex in its active trade lattice:

$$P_A(r) = \text{conv}(P_A(r)) \cap L_A(r).$$

- (iii) **Taylor-slice correspondence.** The cellular chain complex of the compressed quotient is the dehomogenization, at $x_i = 1$, of the principal degree slice of the Bayer–Sturmfels Taylor resolution of the active lattice ideal. The correspondence persists for rank-one local systems after character evaluation.

- (iv) **Rank-two homology.** In active rank two,

$$H_j(I_A(r); \mathbb{Z}) = 0 \quad (j \geq 3),$$

and $H_2(I_A(r); \mathbb{Z})$ is free abelian. Moreover $\beta_1 \in \{2, 3\}$.

(v) **Rank-two fundamental groups.** A nondegenerate parallelogram forces

$$\pi_1(I_A(r)) \cong \mathbb{Z}^2.$$

Without a nondegenerate parallelogram the fundamental group is free of rank β_1 .

- (vi) **Complete rank-two homotopy classification.** The parallelogram-free branch is a wedge of 2 or 3 circles; the parallelogram branch is a torus with a finite wedge of 2-spheres.
- (vii) **Explicit realizations.** Every allowed homotopy type occurs in a completion-total scalar system. A single rank-two system can undergo a residual-driven bifurcation from $S^1 \vee S^1$ to T^2 , and a second family realizes $T^2 \vee \bigvee^m S^2$ for every $m \geq 0$.

Novelty boundary. The topological and algebraic tools used below are classical. We use an equivariant nerve theorem in the form of Fernós–Futer–Hagen [4]; the Taylor and lattice-module machinery is due to Bayer–Sturmfels [2]; the codimension-two projective-dimension bound and minimal syzygy structure come from Peeva–Sturmfels [6]; planar convex-lattice nerves are controlled by Alon–Kalai–Matoušek–Meshulam [1]; and the Laurent projective-freeness input is Swan’s theorem [7]. Freiman’s dimension lemma, Pick’s theorem, and Ehrhart theory are likewise classical [8, 3]. The contribution here is the completion-specific reduction from the global shortest-prefix quotient to the optimal fiber, the resulting Taylor-slice bridge, and the rank-two classification derived from that bridge and from the special planar geometry of optimal fibers. No priority claim is made for the classical ingredients in isolation.

2 The pushdown origin of the completion problem

The mathematical results of this paper are self-contained and do not use an automata theorem as a hypothesis. Nevertheless, the operational origin is useful because it explains why minimum completion, rather than arbitrary factorization, is the object being studied.

For the binary balance language

$$\mathcal{L}_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\},$$

the completion-aware 2:1 machine in [figure 1](#) uses center and side states together with a binary work stack. A transition label $a, A/\gamma$ means that input symbol a is consumed and stack top A is replaced by γ . The dashed boxes list operations available only when the right endmarker is the next unread symbol; they are not ordinary ε -moves.

For a prefix w , the scalar residual

$$\rho(w) = N_1(w) - 2N_0(w)$$

is a coordinate for the amount and direction of weighted imbalance. In the general coprime $p:q$ balance law this becomes $qN_1 - pN_0$. The present paper takes the next step: a matrix residual may have several shortest nonnegative completions of the same length, so minimum demand becomes a finite fiber. The geometry of that fiber is the input to the topology studied below.

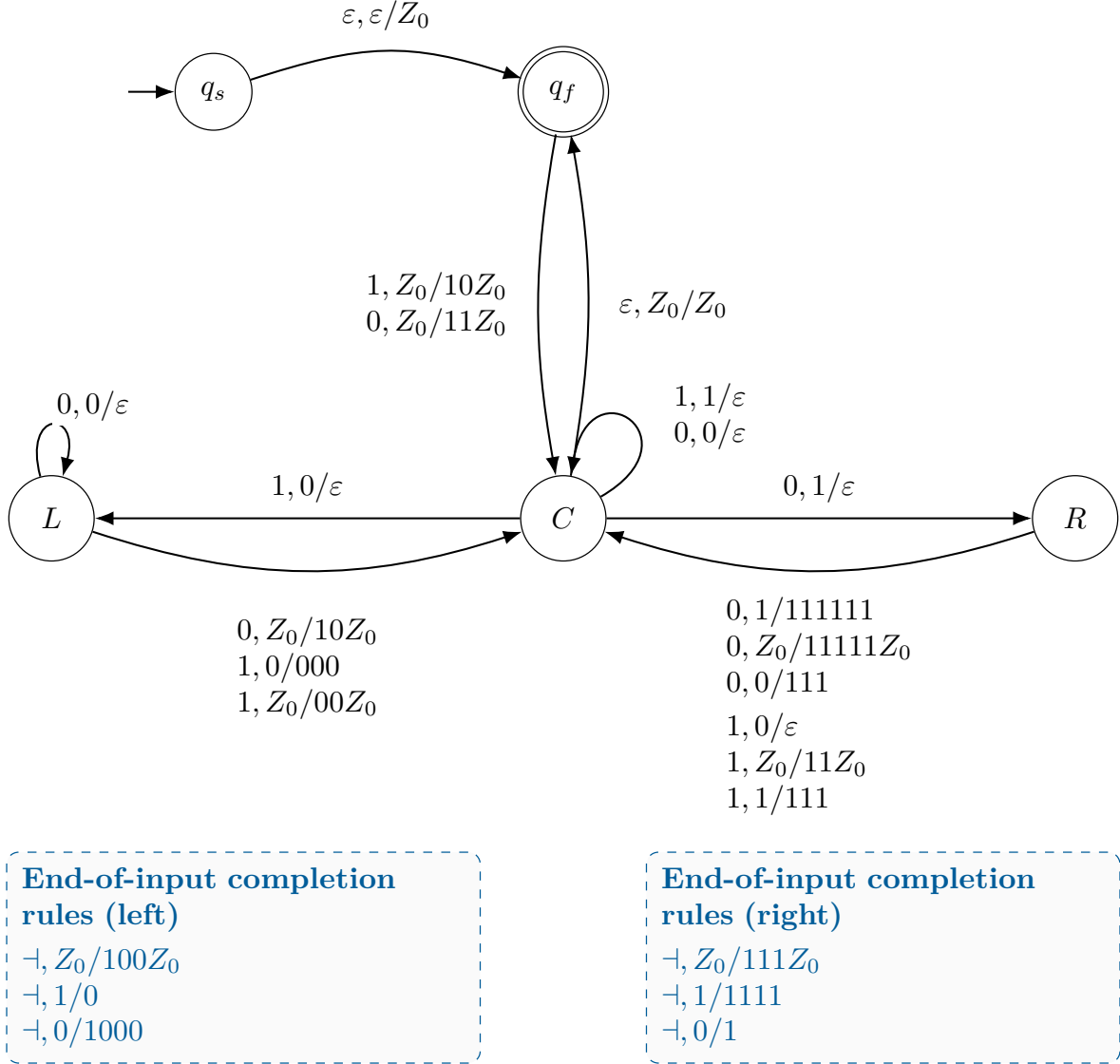


Figure 1: A completion-aware 2:1 DPDA in the empty-word-included presentation. Solid arcs process binary input; dashed boxes list only right-endmarker completion operations. The figure records the operational provenance of shortest-completion semantics; none of the topological theorems below depends on the automaton construction.

3 Minimum-completion geometry

Let $A \in \mathbb{Z}^{k \times d}$. A residual is an element $r \in A\mathbb{Z}^d$ for which the completion equation

$$Ax = -r, \quad x \in \mathbb{N}^d$$

is feasible. A convenient global hypothesis is the following.

Definition 3.1 (Completion totality). The matrix A is *completion-total* if there exists $h \in \mathbb{N}^d$ with every coordinate positive and $Ah = 0$.

If A is completion-total, every lattice residual is completable: if $r = Az$ with $z \in \mathbb{Z}^d$, then $th - z \in \mathbb{N}^d$ for all sufficiently large t , and $A(th - z) = -r$.

For a completable residual r define

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d, Ax = -r\} \quad (1)$$

and the optimal Parikh fiber

$$\mathcal{O}_A(r) = \{x \in \mathbb{N}^d : Ax = -r, \mathbf{1}^\top x = \ell_A(r)\}. \quad (2)$$

The set is finite because all of its points have the same coordinate sum.

The equal-length kernel is

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix} = \{g \in \mathbb{Z}^d : Ag = 0, \mathbf{1}^\top g = 0\}. \quad (3)$$

Differences of optimal completions lie in K_A .

3.1 The shortest-prefix down-set

For $x \in \mathbb{N}^d$ put

$$B_x = \{u \in \mathbb{N}^d : 0 \leq u \leq x\},$$

where inequalities are coordinatewise, and define

$$\mathcal{C}_A(r) = \downarrow \mathcal{O}_A(r) = \bigcup_{x \in \mathcal{O}_A(r)} B_x. \quad (4)$$

Proposition 3.2 (Exact geodesic-prefix criterion). *For $u \in \mathbb{N}^d$,*

$$u \in \mathcal{C}_A(r) \iff \ell_A(r + Au) = \ell_A(r) - \mathbf{1}^\top u.$$

Thus $\mathcal{C}_A(r)$ is exactly the set of Parikh vectors that occur as prefixes of shortest completions of r .

Proof. If $u \leq x$ for some $x \in \mathcal{O}_A(r)$, then $x - u$ completes $r + Au$ and has length $\ell_A(r) - \mathbf{1}^\top u$. A strictly shorter completion would combine with u to contradict optimality of x .

Conversely, if the displayed equality holds and y is a shortest completion of $r + Au$, then $u + y$ completes r and has length $\ell_A(r)$. Hence $u + y \in \mathcal{O}_A(r)$ and $u \in \mathcal{C}_A(r)$. $\square$

Give every finite down-set $D \subset \mathbb{N}^d$ its standard cubical realization $|D|$: a J -oriented elementary cube based at u is present when $u + \mathbf{1}_J \in D$. The down-set condition supplies all faces automatically.

Two equally oriented cells in $|\mathcal{C}_A(r)|$ are identified when their base vertices differ by an element of K_A . We write

$$I_A(r) = |\mathcal{C}_A(r)| / K_A \quad (5)$$

for the resulting filled shortest-prefix quotient. By [theorem 3.2](#), its vertices are precisely shortest-path residual states, and its higher cubes record commuting one-letter shortest moves.

4 Active trades and optimal-fiber lattice geometry

The full kernel K_A can contain directions that never occur inside the optimal fiber for a fixed residual. The correct residual-dependent lattice is smaller.

Definition 4.1 (Active trade lattice). Fix a completable residual r . Define

$$L_A(r) = \langle x - y : x, y \in \mathcal{O}_A(r) \rangle_{\mathbb{Z}} \subseteq K_A. \quad (6)$$

For $x_0 \in \mathcal{O}_A(r)$ define the normalized optimal fiber

$$P_A(r) = \mathcal{O}_A(r) - x_0 \subseteq L_A(r). \quad (7)$$

Its active rank is $\text{rank}_{\mathbb{Z}} L_A(r)$.

Lemma 4.2 (Active collision lemma). *If $u, v \in \mathcal{C}_A(r)$ and $u - v \in K_A$, then*

$$u - v \in \mathcal{O}_A(r) - \mathcal{O}_A(r) \subseteq L_A(r).$$

Proof. The condition $u - v \in K_A$ implies

$$Au = Av, \quad \mathbf{1}^\top u = \mathbf{1}^\top v.$$

Hence u and v reach the same residual at the same geodesic depth. Choose a shortest completion y of that common residual. By [theorem 3.2](#), both $u + y$ and $v + y$ lie in $\mathcal{O}_A(r)$. Their difference is $u - v$. $\square$

Proposition 4.3 (Active-fiber saturation). *For every $x_0 \in \mathcal{O}_A(r)$,*

$$\mathcal{O}_A(r) = (x_0 + L_A(r)) \cap \mathbb{N}^d. \quad (8)$$

Consequently

$$P_A(r) = \text{conv}(P_A(r)) \cap L_A(r). \quad (9)$$

Proof. The inclusion $\mathcal{O}_A(r) \subseteq (x_0 + L_A(r)) \cap \mathbb{N}^d$ follows from the definition of $L_A(r)$. Conversely, if $x = x_0 + \lambda \in \mathbb{N}^d$ with $\lambda \in L_A(r) \subseteq K_A$, then

$$Ax = Ax_0 = -r, \quad \mathbf{1}^\top x = \mathbf{1}^\top x_0 = \ell_A(r),$$

so $x \in \mathcal{O}_A(r)$.

For the second statement, put $P = P_A(r)$. Equation (8) says

$$P = L_A(r) \cap \{z \in \mathbb{R}^d : z_i \geq -(x_0)_i \ \forall i\}.$$

The right-hand real region is convex. Since $\text{conv}(P)$ lies inside it, every lattice point of $\text{conv}(P)$ belongs to P . $\square$

Thus every normalized optimal fiber is a finite lattice-convex configuration in its active lattice. This constraint is decisive in active rank two.

5 Optimal-fiber nerve compression

Let

$$X = |\mathcal{C}_A(r)|, \quad L = L_A(r).$$

Form the periodic saturation

$$\tilde{X} = \bigcup_{\lambda \in L} (X + \lambda). \quad (10)$$

By [theorem 4.2](#), the quotient $\tilde{X}/L$ is canonically the same cell quotient as $I_A(r)$.

Lemma 5.1 (Contractible translate intersections). *Every nonempty finite intersection*

$$\bigcap_{\lambda \in F} (X + \lambda), \quad F \subset L \text{ finite},$$

is contractible.

Proof. Work first on vertices. Put $C = \mathcal{C}_A(r)$ and

$$D_F = \bigcap_{\lambda \in F} (C + \lambda).$$

Assume $D_F \neq \emptyset$ and set

$$m_i(F) = \max_{\lambda \in F} \lambda_i.$$

Every $u \in D_F$ satisfies $u \geq m(F)$ coordinatewise: for each coordinate choose a translate attaining the maximum. If $0 \leq z \leq u - m(F)$, then for every $\lambda \in F$,

$$0 \leq z + m(F) - \lambda \leq u - \lambda.$$

Since $u - \lambda \in C$ and C is a down-set, $z + m(F) - \lambda \in C$. Thus $D_F - m(F)$ is a finite coordinate down-set in $\mathbb{N}^d$. Its cubical realization is a union of coordinate boxes anchored at the origin and is therefore contractible. Translating back proves the claim. $\square$

Choose $x_0 \in \mathcal{O}_A(r)$ and write $P = P_A(r)$. Define the periodic simplicial complex

$$\mathcal{N}(P) = \{F \subseteq L : F \text{ finite and } \exists c \in L \text{ with } c - F \subseteq P\}. \quad (11)$$

Equivalently, $\mathcal{N}(P)$ is the nerve of the family of finite convex-lattice sets

$$\{\lambda + P : \lambda \in L\}.$$

The lattice L acts by translation on $\mathcal{N}(P)$.

Theorem 5.2 (Optimal-Fiber Nerve Compression). *For every completable residual r ,*

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r). \quad (12)$$

The homotopy type of the global shortest-prefix quotient is therefore determined by the finite optimal fiber and its translation lattice.

Proof. The translates $\{X + \lambda\}_{\lambda \in L}$ cover $\tilde{X}$ and are permuted by the L -action. By [theorem 5.1](#), every nonempty finite intersection is contractible. The equivariant simplicial nerve theorem [[4](#), Theorem 2.8] gives an L -equivariant homotopy equivalence between $\tilde{X}$ and the nerve of this cover.

It remains to identify that nerve with (11). A finite $F \subset L$ spans a simplex exactly when there exists a common vertex u with

$$u - \lambda \in \mathcal{C}_A(r) \quad (\lambda \in F).$$

All points $u - \lambda$ have the same residual and the same geodesic depth because their pairwise differences lie in $L \subseteq K_A$. Choose a shortest suffix y from that common residual. Then

$$c = u + y$$

satisfies $c - \lambda \in \mathcal{O}_A(r)$ for every $\lambda \in F$. Conversely, if $c - F \subseteq \mathcal{O}_A(r)$ then c belongs to every translated shortest-prefix set. After translating by x_0 , this is exactly the simplex criterion (11).

Finally, the L -action is free and cellular, so passage to quotients preserves the equivariant homotopy equivalence. Using [theorem 4.2](#), $\tilde{X}/L$ is $I_A(r)$. $\square$

Remark 5.3 (Rank-one rigidity). If $\text{rank } L_A(r) = 1$, lattice-convexity forces $P_A(r)$ to be an interval of consecutive points in a rank-one lattice. If the fiber is nontrivial, $\mathcal{N}(P_A(r))$ is contractible and its translation quotient is S^1 . Thus the first non-rigid regime is active rank two.

6 Finite presentations and first homology

The compressed quotient has one vertex orbit. Its 2-skeleton can be read directly from differences in the optimal fiber.

Proposition 6.1 (Finite fundamental-group presentation). *Assume $0 \in P = P_A(r)$. Then $\pi_1(I_A(r))$ is isomorphic to*

$$G(P) = \langle X_p \ (p \in P), \ X_0 = 1 \mid X_p^{-1}X_q = X_s^{-1}X_t \text{ whenever } q - p = t - s \rangle. \quad (13)$$

Proof. In $\mathcal{N}(P)/L$, oriented edge orbits are indexed by differences $d = q - p \in P - P$. Write the corresponding edge generator as a_d , with $a_{-d} = a_d^{-1}$. Every triple $p, q, s \in P$ supplies the triangular relation

$$a_{q-p}a_{s-q} = a_{s-p}.$$

Set $X_p = a_p$ for $p \in P$ and $X_0 = 1$. The triangle on $0, p, q$ gives

$$a_{q-p} = X_p^{-1}X_q.$$

Two edges represent the same edge orbit exactly when their differences agree, which gives precisely the relations in (13). Higher simplices do not affect the fundamental group. $\square$

There is a canonical displacement homomorphism

$$\delta : G(P) \longrightarrow L, \quad \delta(X_p) = p. \quad (14)$$

It is surjective because $L = \langle P - P \rangle = \langle P \rangle$.

Corollary 6.2 (Parallelogram-commutator principle). *If*

$$p, \quad p + u, \quad p + v, \quad p + u + v \in P,$$

then the corresponding difference generators satisfy

$$[a_u, a_v] = 1.$$

Proof. The two triangles give

$$a_u a_v = a_{u+v} = a_v a_u.$$

$\square$

6.1 Freiman dimension and first homology

Let F be a field of characteristic 0. A normalized Freiman 2-homomorphism on P is a function $f : P \rightarrow F$ with $f(0) = 0$ such that

$$p + q = s + t \implies f(p) + f(q) = f(s) + f(t).$$

Let $\dim_F(P)$ denote the dimension of this vector space. This agrees with the usual normalized Freiman-dimension convention.

Proposition 6.3 (First homology as the Freiman shadow). *For every finite normalized fiber P ,*

$$\beta_1(\mathcal{N}(P)/L) = \dim_F(P).$$

Proof. Abelianizing (13) replaces X_p by additive variables x_p and the defining relations by

$$x_q - x_p = x_t - x_s \quad \text{whenever} \quad q - p = t - s.$$

Equivalently,

$$x_p + x_t = x_q + x_s \quad \text{whenever} \quad p + t = q + s.$$

After $x_0 = 0$, this is exactly the vector space of normalized Freiman homomorphisms. Tensoring H_1 with F gives the stated equality of ranks. $\square$

Theorem 6.4 (Rank-two first-Betti bound). *If $\text{rank } L_A(r) = 2$, then*

$$\beta_1(I_A(r)) \in \{2, 3\}.$$

Proof. Put $P = P_A(r)$, $m = |P|$, and $Q = \text{conv}(P)$ inside $L \otimes \mathbb{R} \cong \mathbb{R}^2$. By [theorem 4.3](#), $P = Q \cap L$. Let B be the number of lattice points on ∂Q . Pick's theorem and the Ehrhart polynomial give

$$|2Q \cap L| = 4m - B - 3 \leq 4m - 6. \quad (15)$$

Since $P + P \subseteq 2Q \cap L$,

$$|P + P| \leq 4m - 6.$$

If $d_F = \dim_F(P)$, Freiman's dimension lemma [8, Chapter 5] gives

$$|P + P| \geq (d_F + 1)m - \binom{d_F + 1}{2}.$$

For $d_F \geq 4$ and $m \geq 5$, the right side is at least $5m - 10 > 4m - 6$. If $m \leq 4$, then trivially $d_F \leq m - 1 \leq 3$. Hence $d_F \leq 3$.

On the other hand, the two coordinate linear functionals on the full-rank planar lattice L give two independent normalized Freiman homomorphisms, so $d_F \geq 2$. Apply [theorem 6.3](#) and [theorem 5.2](#). $\square$

7 The optimal-fiber Taylor slice

The nerve model has a second description in commutative algebra. This description is classical on the lattice-ideal side; the point here is that the minimum-completion quotient selects a canonical principal slice.

Fix a field $\mathbb{k}$, put

$$S = \mathbb{k}[x_1, \dots, x_d],$$

and let $L = L_A(r) \subseteq \mathbb{Z}^d$. Because $L \subseteq K_A$, every $\lambda \in L$ has coordinate sum 0, hence

$$L \cap \mathbb{N}^d = \{0\}.$$

Define the lattice ideal

$$I_L = \langle x^{u^+} - x^{u^-} : u \in L \rangle \subseteq S. \quad (16)$$

Give S the $\mathbb{Z}^d/L$ grading $\deg(x^a) = a + L$. For $\alpha \in \mathbb{N}^d/L$ write

$$\text{fiber}(\alpha) = \{x^a : a \in \mathbb{N}^d, a + L = \alpha\}.$$

Choose $x_0 \in \mathcal{O}_A(r)$ and set

$$\alpha = x_0 + L.$$

By [theorem 4.3](#),

$$\text{fiber}(\alpha) \text{ has exponent set exactly } \mathcal{O}_A(r). \quad (17)$$

For each degree β , let $E_i(\beta)$ be the family of i -element subsets of $\text{fiber}(\beta)$ with monomial gcd equal to 1. Bayer–Sturmfels identify these sets as the homogeneous basis of the lattice-ideal Taylor resolution and give the differential explicitly [2, Proposition 3.10].

Write $\beta \preceq \alpha$ when $\alpha - \beta$ is the degree of a monomial.

Theorem 7.1 (Optimal-Fiber Taylor Slice). *Let*

$$Q(P) = \mathcal{N}(P_A(r))/L_A(r).$$

For every $j \geq 0$ there is a canonical basis correspondence

$$C_j(Q(P); \mathbb{k}) \cong \bigoplus_{\beta \preceq \alpha} \mathbb{k} E_{j+1}(\beta), \quad (18)$$

under which the cellular boundary is the Bayer–Sturmfels Taylor differential specialized by

$$\varepsilon : S \rightarrow \mathbb{k}, \quad x_i \mapsto 1.$$

Equivalently, $C_\bullet(Q(P); \mathbb{k})$ is the dehomogenized principal degree slice of the lattice-ideal Taylor complex.

Proof. Represent a j -simplex orbit of $Q(P)$ by a $(j+1)$ -element subset

$$F = \{u_0, \dots, u_j\} \subseteq \mathcal{O}_A(r).$$

Let $g = \min(u_0, \dots, u_j)$ coordinatewise. Subtracting g gives a gcd-one set

$$\overline{F} = F - g$$

in the fiber of the degree

$$\beta = \alpha - \deg(x^g) \preceq \alpha.$$

If F is replaced by an L -translate, the gcd changes by the same translate and $\overline{F}$ is unchanged. Hence the construction depends only on the simplex orbit.

Conversely, let $I \in E_{j+1}(\beta)$ with $\beta \preceq \alpha$. Choose $g \in \mathbb{N}^d$ with $\alpha = \beta + \deg(x^g)$. Multiplying every monomial of I by x^g gives a subset of $\text{fiber}(\alpha) = \mathcal{O}_A(r)$. A different choice g' differs from g by an element of L and therefore gives the same simplex orbit. The two constructions are inverse.

If one vertex is removed, the remaining set acquires a new common gcd h . Bayer–Sturmfels’ differential is

$$\partial(I) = \sum_{m \in I} \text{sign}(m, I) \gcd(I \setminus \{m\}) [I \setminus \{m\}],$$

where the bracket denotes removal of that common factor. Under $x_i \mapsto 1$, every monomial coefficient becomes 1, leaving exactly the oriented simplicial boundary of the quotient. The shift $E_{j+1} \leftrightarrow C_j$ is the usual one: E_1 is homological degree 0 in the resolution of S/I_L . $\square$

7.1 Principal truncation and character twists

Let $\mathbb{T}_\bullet$ be the full lattice Taylor resolution and $\mathbb{F}_\bullet$ a minimal $\mathbb{Z}^d/L$ -graded free resolution of S/I_L .

Lemma 7.2 (Principal truncation lemma). *The full graded chain-homotopy equivalence between $\mathbb{T}_\bullet$ and $\mathbb{F}_\bullet$ restricts to a chain-homotopy equivalence on the principal degree order ideal*

$$\{\beta : \beta \preceq \alpha\}.$$

Proof. Comparison maps and homotopies may be chosen homogeneous of degree 0. A homogeneous map from a basis summand of degree β to one of degree γ has a monomial coefficient of degree $\beta - \gamma$, so it can be nonzero only when $\gamma \preceq \beta$. Hence every map and homotopy preserves every degree down-set, in particular the principal down-set below α . $\square$

Let $\chi : L \rightarrow \mathbb{k}^\times$ be a character. After a faithfully flat extension of the field if necessary, extend χ to a homomorphism

$$\tilde{\chi} : \mathbb{Z}^d \rightarrow \bar{\mathbb{k}}^\times.$$

Let $\mathbb{k}_\chi$ be the corresponding rank-one local system on $Q(P)$.

Proposition 7.3 (Character-twisted Taylor slice). *After a degree-dependent rescaling of basis vectors, the complex*

$$C_\bullet(Q(P); \mathbb{k}_\chi)$$

is isomorphic to the principal Taylor slice evaluated at

$$x_i = \tilde{\chi}(e_i).$$

Proof. For every $\beta \preceq \alpha$, choose a lift $g_\beta \in \mathbb{N}^d$ with

$$\alpha = \beta + \deg(x^{g_\beta}).$$

Suppose a Taylor boundary term deletes one vertex, produces common gcd exponent h , and passes from degree β to γ . The corresponding face in the periodic nerve differs from the chosen representative in degree γ by the deck translation

$$\tau = h + g_\beta - g_\gamma \in L.$$

After rescaling the degree- β basis by $\tilde{\chi}(g_\beta)$, the evaluated Taylor coefficient becomes

$$\tilde{\chi}(h) \frac{\tilde{\chi}(g_\beta)}{\tilde{\chi}(g_\gamma)} = \tilde{\chi}(\tau) = \chi(\tau),$$

which is exactly the local-system coefficient attached to that face. The signs are the common simplicial incidence signs. $\square$

8 Rank-two homological dimension

Assume from now on that

$$\text{rank}_{\mathbb{Z}} L_A(r) = 2.$$

Peeva–Sturmfels prove the general projective-dimension bound

$$\text{pd}_S(S/I_L) \leq 2^{\text{rank } L} - 1,$$

so in rank two the minimal resolution has no term above homological degree 3 [6].

Lemma 8.1 (Torus-point injectivity). *For every field $\mathbb{k}$ and every character $\chi : L \rightarrow \mathbb{k}^\times$, the third differential of the full minimal resolution, after the corresponding torus-point specialization, is injective.*

Proof. Extend scalars to an algebraic closure and represent χ by a point $\xi \in (\overline{\mathbb{k}}^\times)^d$.

If χ is nontrivial, then some lattice binomial evaluates nonzero at ξ , so ξ lies outside the support of S/I_L . The specialized full free resolution is therefore acyclic.

If χ is trivial, localize at the algebraic torus. For a lattice basis g_1, g_2 of L , the Laurent lattice ideal is generated by

$$x^{g_1} - 1, \quad x^{g_2} - 1.$$

These two independent equations form a regular sequence in the regular Laurent ring. Hence the localized module is a codimension-two complete intersection and its Tor vanishes above degree 2. In both cases the third homology of the specialized full minimal resolution is zero. Since there is no fourth term, the third differential is injective. $\square$

Theorem 8.2 (Rank-Two Homological-Dimension Theorem). *If $\text{rank } L_A(r) = 2$, then*

$$H_j(I_A(r); \mathbb{Z}) = 0 \quad (j \geq 3), \tag{19}$$

and $H_2(I_A(r); \mathbb{Z})$ is free abelian.

Proof. Over any field $\mathbb{k}$, [theorems 7.1](#) and [7.2](#) identify the chain complex of $Q(P)$ up to chain homotopy with the principal slice of a minimal free resolution of length at most 3. Thus homology vanishes above degree 3. By [theorem 8.1](#), the third differential remains injective after restriction to the principal slice, so

$$H_j(Q(P); \mathbb{k}) = 0 \quad (j \geq 3).$$

The same holds for every prime field and for $\mathbb{Q}$. Since $Q(P)$ is finite, the universal coefficient theorem implies (19). Applying the mod- p universal coefficient sequence in degree 3 gives

$$\text{Tor}_1^{\mathbb{Z}}(H_2(Q(P); \mathbb{Z}), \mathbb{F}_p) = 0$$

for every prime p , so $H_2(Q(P); \mathbb{Z})$ has no torsion. Finally use [theorem 5.2](#). $\square$

9 Planar lattice geometry and the fundamental group

By [theorem 4.3](#), a rank-two normalized fiber is a finite lattice-convex set in a rank-two lattice. We now show that a single geometric feature—a nondegenerate parallelogram—separates the abelian and free branches of the fundamental group.

9.1 The parallelogram branch

Lemma 9.1 (Minimal parallelogram lemma). *Let $P \subset L \cong \mathbb{Z}^2$ be finite and lattice-convex. If P contains a nondegenerate lattice parallelogram, then it contains one of normalized area 1 or 2 whose edge directions are primitive.*

Proof. Choose a nondegenerate parallelogram

$$R = \{a, a + u, a + v, a + u + v\} \subseteq P$$

of minimum positive normalized area $D = |\det_L(u, v)|$. If u were not primitive, an interior lattice point on each of the two u -parallel edges would belong to P by lattice-convexity and would produce a smaller parallelogram. Thus u and v are primitive. Hence the boundary of $\text{conv}(R)$ contains exactly four lattice points. Pick's theorem gives

$$I(R) = D - 1$$

interior lattice points.

Let c be the center of R . Reflection in c preserves the lattice and the parallelogram. If there is an interior lattice point $z \neq c$, its reflection $z' = 2c - z$ is another interior lattice point. One of the two parallelograms formed by the pair z, z' and one of the two opposite vertex pairs of R is nondegenerate; it lies strictly inside R and has smaller normalized area, contradicting minimality. Hence R has at most one interior lattice point, so $D - 1 \leq 1$. $\square$

A finite lattice-convex set $X \subset L$ will be called *parity-complete* if it meets every coset of $2L$.

Lemma 9.2 (Observer chain). *If $X \subsetneq P$ are finite lattice-convex subsets of L , there exists $z \in P \setminus X$ such that*

$$\text{conv}(X \cup \{z\}) \cap L = X \cup \{z\}.$$

Iterating gives a chain from X to P by one-point lattice-convex extensions.

Proof. Choose $z \in P \setminus X$ minimizing the positive Euclidean distance to $\text{conv}(X)$. If a further lattice point $y \notin X \cup \{z\}$ belonged to $\text{conv}(X \cup \{z\})$, then $y \in P$ by lattice-convexity and, along a segment from z toward $\text{conv}(X)$, would have strictly smaller distance to $\text{conv}(X)$, a contradiction. $\square$

Lemma 9.3 (Parity propagation). *Suppose $X \subseteq P$ is lattice-convex and parity-complete, and suppose that in $G(P)$ all X_x with $x \in X$ lie in the subgroup generated by commuting elements U, V whose displacements form a basis of L . If z is an observer of X , then X_z lies in the same subgroup, with exponent vector equal to the lattice coordinates of z .*

Proof. Choose $x \in X$ with $x \equiv z \pmod{2L}$ and put $y = (x + z)/2 \in L$. The observer property forces $y \in X$. Since

$$z - y = y - x,$$

relation (13) gives

$$X_y^{-1} X_z = X_x^{-1} X_y, \quad X_z = X_y X_x^{-1} X_y.$$

If $X_x = U^a V^b$ and $X_y = U^c V^d$, commutativity gives

$$X_z = U^{2c-a} V^{2d-b},$$

whose displacement is $2y - x = z$. $\square$

Lemma 9.4 (Determinant-two parity completion). *Let*

$$C_2 = \{(0, 0), (1, 0), (0, 1), (-1, 2), (0, 2)\} \subset \mathbb{Z}^2.$$

If a lattice-convex set $P \supseteq C_2$ contains a point in the missing parity class $(1, 1) + 2\mathbb{Z}^2$, then it contains at least one point of

$$M = \{(1, -1), (1, 1), (-1, 1), (-1, 3)\}.$$

For every $m \in M$, the set $C_2 \cup \{m\}$ is lattice-convex and parity-complete, and the generator X_m is already a word in the generators from C_2 .

Proof. Consider the linear map

$$\Phi(x, y) = (y, 2x + y).$$

It sends $\text{conv}(C_2)$ to the square $[0, 2]^2$. If z lies in the missing parity class, write

$$\Phi(z) = (s, t), \quad A = s - 1, \quad B = t - 1.$$

Then A, B are even and $B - A \equiv 2 \pmod{4}$, so $|A| \neq |B|$ and therefore

$$||A| - |B|| \geq 2.$$

One of the four inequalities

$$B \geq |A| + 2, \quad -B \geq |A| + 2, \quad A \geq |B| + 2, \quad -A \geq |B| + 2$$

holds. In the first case, the point $(1, 3)$ lies in $\text{conv}([0, 2]^2 \cup \{\Phi(z)\})$: indeed, with $\theta = 1/(B - 1)$, write it as a convex combination of $\Phi(z)$ and a point of the top side using $|A| \leq B - 2$. The other three cases are symmetric. The four resulting points are precisely the images under Φ of the four elements of M . Since P is lattice-convex, the corresponding element of M lies in P .

A direct check of the four small polygons shows

$$\text{conv}(C_2 \cup \{m\}) \cap \mathbb{Z}^2 = C_2 \cup \{m\}$$

for each $m \in M$, and the added point supplies the fourth parity class. Finally, in each case there are $p, a, b \in C_2$ with $m = p + b - a$. The relation

$$X_p^{-1}X_m = X_a^{-1}X_b$$

expresses X_m in the subgroup generated by C_2 . □

Theorem 9.5 (Parallelogram-abelian fundamental group). *Let $P = P_A(r)$ have active rank two. If P contains a nondegenerate lattice parallelogram, then*

$$\pi_1(I_A(r)) \cong L_A(r) \cong \mathbb{Z}^2.$$

Proof. By [theorem 9.1](#), choose a minimal parallelogram of normalized area 1 or 2.

If the area is 1, translate one vertex to 0 and let u, v be its primitive edge directions. They form a basis of L , the four vertices represent all four classes in $L/2L$, and [theorem 6.2](#) gives commuting basis generators X_u, X_v . The four-point core is parity-complete. Apply [theorems 9.2](#) and [9.3](#) inductively to reach all of P .

If the area is 2, let the primitive edge directions be a, b . Modulo $2L$ they are equal, so the center

$$v = \frac{a + b}{2}$$

is a lattice point. With $u = a$, the pair u, v is a lattice basis and the five lattice points of the minimal parallelogram have coordinates

$$C_2 = \{0, u, v, 2v - u, 2v\}.$$

The relations

$$X_{2v} = X_v^2, \quad X_{2v-u} = X_v X_u^{-1} X_v, \quad X_u = X_{2v-u}^{-1} X_{2v}$$

force $[X_u, X_v] = 1$. This core represents three parity classes. If P contains only those three classes, [theorems 9.2](#) and [9.3](#) again propagate the basis expressions to all of P . If the fourth class occurs, [theorem 9.4](#) enlarges the core by one already-dependent generator to a parity-complete lattice-convex core, after which the same induction applies.

Thus $G(P)$ is generated by two commuting elements whose displacements are a basis of L . The displacement map (14) is therefore an isomorphism $G(P) \cong L$. Use [theorems 5.2](#) and [6.1](#). □

9.2 The parallelogram-free branch

Call an affine lattice line ℓ *rich* if $|P \cap \ell| \geq 3$. Lattice-convexity implies that $P \cap \ell$ is a consecutive arithmetic progression in the primitive direction of ℓ .

Lemma 9.6 (Line-locality of repeated differences). *If P contains no nondegenerate parallelogram and*

$$q - p = t - s$$

for two distinct ordered pairs of points of P , then the two segments lie on the same affine lattice line.

Proof. If the two segments lie on different parallel lines, the four endpoints form a nondegenerate parallelogram. If the lines are not parallel, equal displacement is impossible. $\square$

A point lying on at least two rich lines will be called *shared*.

Lemma 9.7 (Transverse determinant lemma). *Let P be lattice-convex and parallelogram-free. Suppose*

$$0, u, 2u, -v, v \in P$$

with primitive, nonparallel u, v . Then

$$|\det_L(u, v)| = 1.$$

Proof. Choose coordinates $u = (1, 0)$ and $v = (a, b)$ with $b > 0$ and $\gcd(a, b) = 1$. The triangle

$$T = \text{conv}\{-v, v, 2u\}$$

lies in $\text{conv}(P)$, hence every lattice point of T lies in P .

For $b \geq 3$, the intersection $T \cap (2u - T)$ has, in the u, v coordinate system, the form

$$\{xu + yv : |x - 1| + 2|y| \leq 1\}.$$

Take the ordinary lattice point $z = (m, 1)$ where m is nearest to $1 + a/b$. Then $y = 1/b$ and

$$|x - 1| = \text{dist}(a/b, \mathbb{Z}) \leq \frac{\lfloor b/2 \rfloor}{b} \leq 1 - \frac{2}{b}.$$

Thus z and $2u - z$ both lie in $T \cap L \subseteq P$. They are not on the u -line, and

$$0 + 2u = z + (2u - z)$$

gives a nondegenerate parallelogram, a contradiction.

If $b = 2$, a lattice shear gives $v = (1, 2)$. Then the four points

$$(1, 2), \quad (0, -1), \quad (0, 0), \quad (1, 1)$$

lie in $T \cap L \subseteq P$ and satisfy

$$(1, 2) + (0, -1) = (0, 0) + (1, 1),$$

again a nondegenerate parallelogram. Hence $b = 1$. $\square$

Lemma 9.8 (Endpoint-bridge exclusion). *Let P be lattice-convex and parallelogram-free. A rich segment of lattice length at least 2 cannot have both endpoints interior to transverse rich lines.*

Proof. Write the rich segment as

$$0, u, 2u, \dots, ku, \quad k \geq 2,$$

and suppose the two endpoints are interior to transverse rich lines, so

$$-v, 0, v \in P, \quad ku - w, ku, ku + w \in P.$$

By [theorem 9.7](#),

$$|\det(u, v)| = |\det(u, w)| = 1.$$

Choose coordinates $u = (1, 0)$, $v = (0, 1)$. Then $w = (a, \varepsilon)$ with $\varepsilon = \pm 1$. Hence at the levels $y = 1$ and $y = -1$ the right transverse line contributes points with horizontal coordinates $k + a$ and $k - a$ (in some order). At least one of these is nonzero. On the same level the left transverse line contributes $(0, \pm 1)$. Convexity therefore forces a horizontal unit edge on that level. The main rich segment contains the horizontal unit edge from $(0, 0)$ to $(1, 0)$. The two equal unit edges form a nondegenerate parallelogram, a contradiction. $\square$

Lemma 9.9 (Rich-line incidence). *If P is lattice-convex and parallelogram-free, then every rich line contains at most two shared points; if it contains two, they are consecutive lattice points on that line.*

Proof. A point cannot be interior to two transverse rich lines: the four neighboring points have the same midpoint and form a nondegenerate parallelogram. Thus if a shared point p is interior to a rich line D , it is an endpoint of every transverse rich line through it. Let u be the primitive direction of D and let the transverse line contain $p, p + v, p + 2v$. If D extended two steps from p in the same direction, then $p + 2u, p + 2v \in P$ and lattice-convexity would give

$$p + u + v = \frac{(p + 2u) + (p + 2v)}{2} \in P,$$

producing the parallelogram $p, p + u, p + v, p + u + v$. Hence a shared interior point can be at most one lattice step from each endpoint of D .

A point cannot be an endpoint of two transverse rich lines either: the two two-step extensions yield the same midpoint argument. Consequently, if a rich line had three shared points, its middle shared point would force the line to consist of exactly three consecutive points, while the two endpoints would be interior to transverse rich lines, contradicting [theorem 9.8](#). If two shared points were nonconsecutive, neither could be an interior point of the rich line, so both would be endpoints and both would be interior to transverse rich lines, again contradicting [theorem 9.8](#). $\square$

Theorem 9.10 (Parallelogram-free homotopy theorem). *Let $P = P_A(r)$ have active rank two and contain no nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq \bigvee^{\beta_1(I_A(r))} S^1. \quad (20)$$

In particular,

$$\pi_1(I_A(r)) \cong F_{\beta_1(I_A(r))}.$$

Proof. By [theorem 9.6](#), every nontrivial positive-dimensional face identification in a fundamental simplex on P is supported inside the simplex on $P \cap \ell$ for some rich line ℓ . Let R be the union of these rich-line simplices together with the full 0-skeleton. Then the quotient $Q(P)$ is the pushout

obtained from the full simplex $\Delta(P)$ by replacing R by its translation quotient $\overline{R}$. Since $\Delta(P)$ is contractible and $R \hookrightarrow \Delta(P)$ is a cofibration,

$$Q(P) \simeq \text{Cofib}(R \rightarrow \overline{R}).$$

Distinct rich-line simplices meet only at shared vertices. By [theorem 9.9](#), each rich line contains at most two such vertices, and if there are two they are consecutive. Hence R deformation retracts to a finite incidence graph. The quotient of a single rich-line simplex is the rank-one completion quotient, hence a circle; no positive-dimensional face is identified across distinct rich lines by [theorem 9.6](#). Thus $\overline{R}$ retracts to a wedge of one circle for each rich line.

Choose a spanning forest in the incidence graph. In the cellular mapping cone of $R \rightarrow \overline{R}$, every non-tree incidence edge gives one 2-cell relation. Because the two shared points on the corresponding rich line are consecutive, its attaching word traverses that rich-line circle generator with coefficient ± 1 . Choosing fundamental cycles with respect to the forest makes that circle generator unique to the corresponding non-tree relation. Successive elementary 2-cell/1-cell cancellations remove all 2-cells. The mapping cone therefore collapses to a finite graph. Its first Betti number is the first Betti number of $Q(P)$, so the graph is homotopy equivalent to the wedge in (20). Use [theorem 5.2](#). $\square$

Combining [theorems 6.4, 9.5 and 9.10](#) gives the complete rank-two fundamental-group list

$$F_2, \quad F_3, \quad \mathbb{Z}^2.$$

10 The parallelogram branch and exact homotopy type

Assume $P = P_A(r)$ contains a nondegenerate parallelogram. By [theorem 9.5](#),

$$\pi_1(Q(P)) \cong L \cong \mathbb{Z}^2.$$

Hence the periodic nerve

$$\tilde{Q} := \mathcal{N}(P)$$

is the universal cover of $Q(P)$.

Let

$$R_L = \mathbb{Z}[L] \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}], \quad M = H_2(\tilde{Q}; \mathbb{Z}).$$

The next result is the module-theoretic step that turns the homology classification into a homotopy classification.

Lemma 10.1 (Universal-cover higher homology). *For active rank two,*

$$H_j(\mathcal{N}(P); \mathbb{Z}) = 0 \quad (j \geq 3).$$

Proof. The complex $\mathcal{N}(P)$ is the nerve of the family of translates $\{\lambda + P\}_{\lambda \in L}$. Since $P = \text{conv}(P) \cap L$, these are convex lattice sets in $L \cong \mathbb{Z}^2$. Alon–Kalai–Matoušek–Meshulam prove that the nerve of every finite family of convex lattice sets in $\mathbb{Z}^d$ is $(2^d - 1)$ -collapsible [1]; in dimension two it is therefore 3-Leray. Any finite cellular cycle in $\mathcal{N}(P)$ is supported on a finite induced subcomplex, so homology in degree at least 3 vanishes. $\square$

Proposition 10.2 (Projectivity of the π_2 module). *In the parallelogram branch, M is a finitely generated projective R_L -module.*

Proof. The universal-cover cellular chains are finitely generated free R_L -modules in each degree, and R_L is Noetherian, so M is finitely generated.

Let $\mathfrak{m}$ be a maximal ideal of R_L and let $k(\mathfrak{m})$ be its residue field. Since the group elements of L are units in R_L , their images define a character

$$\chi_{\mathfrak{m}} : L \rightarrow k(\mathfrak{m})^{\times}.$$

For the free R_L -chain complex $C_{\bullet}(\tilde{Q})$, the hyperhomology spectral sequence gives

$$E_{p,q}^2 = \mathrm{Tor}_p^{R_L}(k(\mathfrak{m}), H_q(\tilde{Q}; \mathbb{Z})) \implies H_{p+q}(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}).$$

Because $\tilde{Q}$ is simply connected, $H_1(\tilde{Q}) = 0$, and by [theorem 10.1](#) only the rows $q = 0, 2$ can be nonzero. The $q = 0$ row is the Koszul homology of the two augmentation elements $s - 1, t - 1$ after evaluation at the residue character; it vanishes in degrees above 2, and for a nontrivial character it is acyclic.

On the other hand, [theorems 7.3](#) and [8.1](#) give

$$H_3(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}) = 0.$$

No differential can kill or create the term $E_{1,2}^2$, so

$$\mathrm{Tor}_1^{R_L}(k(\mathfrak{m}), M) = 0$$

for every maximal ideal $\mathfrak{m}$. Localizing, the finite $R_{L,\mathfrak{m}}$ -module $M_{\mathfrak{m}}$ has vanishing first Tor with its residue field. A minimal local presentation and Nakayama's lemma therefore show that $M_{\mathfrak{m}}$ is free. Thus M is locally free at every maximal ideal, hence finitely generated projective. $\square$

Theorem 10.3 (Parallelogram-branch homotopy theorem). *Let $P = P_A(r)$ have active rank two and contain a nondegenerate lattice parallelogram. Put*

$$b = \beta_2(I_A(r)).$$

Then $b \geq 1$ and

$$I_A(r) \simeq T^2 \vee \bigvee^{b-1} S^2. \tag{21}$$

Proof. By [theorem 9.5](#), $\pi_1(Q(P)) \cong \mathbb{Z}^2$. The Hopf exact sequence therefore gives a surjection

$$H_2(Q(P); \mathbb{Z}) \twoheadrightarrow H_2(\mathbb{Z}^2; \mathbb{Z}) \cong \mathbb{Z},$$

so $b \geq 1$.

By [theorem 10.2](#), M is finitely generated projective over

$$R_L \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}].$$

Swan's Laurent polynomial theorem [\[7\]](#) implies that every finitely generated projective module over this ring is free, so

$$M \cong R_L^q$$

for some $q \geq 0$.

To determine q , take any nontrivial character over a field. In the spectral sequence used above, the $q = 0$ row is acyclic and the only nonzero homology of the local-system complex can occur in degree 2. Hence

$$\dim H_2(Q(P); \mathbb{k}_{\chi}) = q.$$

The Euler characteristic of a rank-one local-system chain complex is the ordinary Euler characteristic. By [theorem 8.2](#),

$$\chi(Q(P)) = 1 - 2 + b = b - 1.$$

Thus $q = b - 1$, and

$$\pi_2(Q(P)) \cong H_2(\tilde{Q}) \cong R_L^{b-1}$$

by Hurewicz on the simply connected universal cover.

Let

$$Y = T^2 \vee \bigvee^{b-1} S^2.$$

Choose a map $T^2 \rightarrow Q(P)$ inducing the isomorphism on π_1 ; the torus 2-cell extends because the two basis loops commute. Choose maps of the 2-spheres representing an R_L -basis of $\pi_2(Q(P))$. Together they define

$$f : Y \rightarrow Q(P)$$

inducing isomorphisms on π_1 and on the R_L -module π_2 . Lift to universal covers. Both universal covers are simply connected; both have zero homology above degree 2 by [theorem 10.1](#), and the lifted map is an isomorphism in degree 2. Hence it is an integral homology equivalence. The simply connected homological Whitehead theorem [5] makes the lifted map a homotopy equivalence, and therefore so is f .

Finally use [theorem 5.2](#). □

11 Complete active-rank-two classification

We can now combine the geometric and algebraic branches.

Theorem 11.1 (Complete rank-two homotopy classification). *Let r be a completable residual with*

$$\text{rank } L_A(r) = 2.$$

Then exactly one of the following holds.

(a) *The normalized optimal fiber $P_A(r)$ is nondegenerate-parallelogram-free. Then*

$$I_A(r) \simeq \bigvee^{\beta_1} S^1, \quad \beta_1 \in \{2, 3\}.$$

(b) *The normalized optimal fiber contains a nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq T^2 \vee \bigvee^{\beta_2-1} S^2, \quad \beta_2 \geq 1.$$

Consequently the possible active-rank-two homotopy types are precisely

$$S^1 \vee S^1, \quad S^1 \vee S^1 \vee S^1, \quad T^2 \vee \bigvee^m S^2 \quad (m \geq 0). \quad (22)$$

Proof. Apply [theorems 6.4](#), [9.10](#) and [10.3](#). The two cases are complementary by definition. □

Corollary 11.2 (Integral homology profiles). *In active rank two the integral homology is exactly one of*

$$\begin{aligned} H_0 &\cong \mathbb{Z}, & H_1 &\cong \mathbb{Z}^2, & H_j &= 0 \quad (j \geq 2), \\ H_0 &\cong \mathbb{Z}, & H_1 &\cong \mathbb{Z}^3, & H_j &= 0 \quad (j \geq 2), \end{aligned}$$

or

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_2 \cong \mathbb{Z}^b, \quad H_j = 0 \quad (j \geq 3), \quad b \geq 1.$$

12 Explicit scalar realizations

The classification is not merely formal: all allowed types occur in four-letter scalar completion systems.

12.1 A fixed-system residual bifurcation

For an integer $d \geq 5$, define

$$\omega_d = (1 - 2d, 1 - d, 1, 1 + d). \quad (23)$$

This row vector is completion-total; for example

$$h_d = (1, 1, 2d - 3, 1) > 0, \quad \omega_d h_d^\top = 0.$$

The equal-length kernel is

$$K_{\omega_d} = \mathbb{Z}u \oplus \mathbb{Z}v, \quad u = (1, -2, 1, 0), \quad v = (2, -3, 0, 1). \quad (24)$$

(We write the first basis vector as u below.)

Theorem 12.1 (Fixed-system rank-two bifurcation). *For the single scalar system (23), define*

$$r_W = 5d - 4, \quad r_T = 3d - 4.$$

Then

$$\ell_{\omega_d}(r_W) = \ell_{\omega_d}(r_T) = 4,$$

and in both cases the active lattice is the full equal-length kernel $K_{\omega_d} \cong \mathbb{Z}^2$. Nevertheless,

$$I_{\omega_d}(r_W) \simeq S^1 \vee S^1, \quad I_{\omega_d}(r_T) \simeq T^2.$$

Proof. For any $x \in \mathbb{N}^4$ of length n put

$$s = x_2 + 2x_3 + 3x_4.$$

Then

$$\omega_d \cdot x = (1 - 2d)n + ds.$$

For both residuals the completion equation implies

$$n \equiv 4 \pmod{d}.$$

Since $d \geq 5$ and length-4 solutions exist, both minimum lengths equal 4.

For r_W , setting $n = 4$ reduces the equation to $s = 3$. Its complete set of nonnegative solutions is

$$\mathcal{O}_W = \{(1, 3, 0, 0), (2, 1, 1, 0), (3, 0, 0, 1)\}. \quad (25)$$

Relative to the first point, the two nonzero differences are

$$u = (1, -2, 1, 0), \quad v = (2, -3, 0, 1),$$

which form the kernel basis. The normalized fiber has three affinely independent lattice points and no nondegenerate parallelogram, so [theorem 11.1](#) gives $S^1 \vee S^1$.

For r_T , the length-4 equation is $s = 5$ and the complete optimal fiber is

$$\mathcal{O}_T = \{(0, 3, 1, 0), (1, 1, 2, 0), (1, 2, 0, 1), (2, 0, 1, 1)\}. \quad (26)$$

Relative to the first point, let

$$g = u, \quad h = v - u = (1, -1, -1, 1).$$

Then the four normalized points are

$$\{0, g, h, g + h\},$$

a fundamental lattice parallelogram. Its quotient has Euler characteristic 0, so [theorems 8.2](#) and [9.5](#) give $\beta_2 = 1$, and [theorem 10.3](#) gives T^2 . $\square$

Thus even the matrix, the full equal-length kernel, the active lattice, its rank, and the minimum completion length can be held fixed while the global topology changes. The missing datum is the residual-dependent optimal-fiber translation geometry.

12.2 Unbounded second homology at fixed rank two

For $m \geq 2$ put

$$\alpha = 2m + 1$$

and define

$$\omega_m = (1 - m\alpha, 1 - (m - 1)\alpha, 1, 1 + \alpha), \quad (27)$$

$$r_m = \alpha m(m - 1) - 2m. \quad (28)$$

The system is completion-total; for example

$$h_m = (1, 1, (2m - 2)\alpha - 3, 1) > 0$$

satisfies $\omega_m h_m^\top = 0$.

Theorem 12.2 (Unbounded rank-two sphere family). *For every $m \geq 2$,*

$$\ell_{\omega_m}(r_m) = 2m.$$

The complete optimal fiber is

$$\mathcal{O}_m = \{A_0, \dots, A_m, B, C\}, \quad (29)$$

where

$$A_t = (t, m - t, m - t, t), \quad 0 \leq t \leq m,$$

$$B = (0, m + 1, 0, m - 1), \quad C = (m - 1, 0, m + 1, 0).$$

Its active lattice equals the full equal-length kernel and has rank two. Moreover

$$I_{\omega_m}(r_m) \simeq T^2 \vee \bigvee^{m-2} S^2, \quad (30)$$

so

$$\beta_2(I_{\omega_m}(r_m)) = m - 1.$$

Proof. All four weights in (27) are congruent to 1 modulo α . Hence every completion of r_m with length n satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m+1}.$$

The vectors in (29) have length $2m$, so $\ell = 2m$.

At length $2m$, eliminating the common congruence term gives

$$x_2 + mx_3 + (m+1)x_4 = m(m+1).$$

Modulo m ,

$$x_2 + x_4 \equiv 0 \pmod{m},$$

and $0 \leq x_2 + x_4 \leq 2m$. Thus $x_2 + x_4$ is 0, m , or $2m$. The three cases give respectively C , the family A_t , and B , proving (29).

Set

$$g = (1, -1, -1, 1), \quad h = (0, 1, -m, m-1).$$

Every equal-length kernel vector is uniquely $ag + bh$: indeed the equations $\mathbf{1}^\top z = 0$ and $\omega_m \cdot z = 0$ imply

$$z_3 = -(m+1)z_1 - mz_2, \quad z_4 = mz_1 + (m-1)z_2,$$

which corresponds to $a = z_1$, $b = z_1 + z_2$. Hence g, h form a basis of the full kernel. Relative to A_0 ,

$$A_t - A_0 = tg, \quad B - A_0 = h, \quad C - A_0 = (m-1)g - h.$$

Thus in active coordinates

$$P_m = \{(t, 0) : 0 \leq t \leq m\} \cup \{(0, 1), (m-1, -1)\}. \quad (31)$$

It contains a nondegenerate parallelogram, for example the four points corresponding to $(0, 1)$, $(m-1, -1)$, $(0, 0)$, and $(m-1, 0)$.

It remains to compute β_2 . Count simplex orbits in the finite quotient using (31). The horizontal-only part is the rank-one quotient and has Euler characteristic 0. The total alternating contribution of faces containing exactly one of the two off-line points, excluding the already-counted singleton vertex, is -1 for each off-line point. Faces containing both off-line points have total alternating contribution 0. The only cross-identifications between the two one-off-line families occur for the m edge pairs

$$\{(0, 1), (t, 0)\} \sim \{(m-1, -1), (m-1-t, 0)\}, \quad 0 \leq t \leq m-1.$$

No face of size at least 3 has such a cross-identification because the two off-line levels are unique. Each edge identification raises the Euler characteristic by 1. Therefore

$$\chi(I_{\omega_m}(r_m)) = m - 2.$$

The parallelogram branch has $\beta_1 = 2$, and theorem 8.2 gives no homology above degree 2. Hence

$$m - 2 = 1 - 2 + \beta_2, \quad \beta_2 = m - 1.$$

Apply theorem 10.3 to obtain (30). □

Corollary 12.3 (Fixed-rank unbounded complexity). *There exist completion-total four-letter scalar systems with*

$$\text{rank } K = \text{rank } L = 2, \quad \pi_1 \cong \mathbb{Z}^2,$$

for which β_2 is arbitrarily large.

12.3 A free-rank-three example

Example 12.4 (The F_3 branch). Take

$$\omega = (-11, -8, -5, 4), \quad r = 25.$$

All weights are 1 modulo 3, so every completion length satisfies $n \equiv 2 \pmod{3}$. Length 2 is impossible because the smallest sum of two weights is $-22 > -25$, while length 5 solutions exist. At length 5 the equation becomes

$$x_2 + 2x_3 + 5x_4 = 10,$$

whose complete nonnegative solution set is

$$\mathcal{O} = \{(0, 0, 5, 0), (0, 3, 1, 1), (1, 1, 2, 1), (3, 0, 0, 2)\}.$$

The active lattice has rank two, and the six unordered pair differences are distinct up to sign. Hence there is no nontrivial repeated-difference relation and no nondegenerate parallelogram. Presentation (13) is therefore free on the three nonzero normalized fiber points. By [theorem 9.10](#),

$$I_\omega(25) \simeq S^1 \vee S^1 \vee S^1.$$

Together, [theorems 12.1](#), [12.2](#) and [12.4](#) realize every homotopy type in (22).

13 Computational consistency checks

The theorems above are symbolic and do not depend on computation. Small exhaustive enumerations were used only as independent checks of the fiber formulas, cell-orbit conventions, and the rank-two dichotomy.

For the fixed-system family (23), direct enumeration for $5 \leq d \leq 20$ recovers exactly the two fibers in equations (25) and (26) and minimum length 4 for the stated residuals. For the unbounded family (27), direct enumeration for $2 \leq m \leq 12$ recovers exactly (29). The first cases are shown in [table 1](#).

m	α	r_m	$ \mathcal{O}_m $	predicted homotopy type
2	5	6	5	T^2
3	7	36	6	$T^2 \vee S^2$
4	9	100	7	$T^2 \vee 2S^2$
5	11	210	8	$T^2 \vee 3S^2$
6	13	378	9	$T^2 \vee 4S^2$

Table 1: Initial cases of the explicit unbounded family. The computations are consistency checks only; [theorem 12.2](#) is proved symbolically.

A separate finite-grid audit of planar lattice-convex fibers was used during discovery to test the parallelogram/free-group dichotomy and the Betti patterns. These experiments motivated the geometric lemmas in [section 9](#); they are not proof premises and no computational assumption enters [theorem 11.1](#).

14 Conclusion

The completion-geometric passage from a unique binary shortest demand to a higher-dimensional optimal fiber introduces topology that is invisible to kernel rank alone. The central reduction of this paper shows that the entire filled shortest-prefix quotient can be compressed to the periodic translation nerve of the finite optimal fiber:

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r).$$

The active lattice determines the translation group, but not the global homotopy type. The missing information lies in the additive and convex geometry of $P_A(r)$.

In active rank two, that principle becomes a complete classification. A parallelogram-free optimal fiber produces a free fundamental group and a wedge of two or three circles. A nondegenerate parallelogram forces the fundamental group to become $\mathbb{Z}^2$, and the remaining topology consists exactly of a finite wedge of 2-spheres attached to a torus. The Taylor-slice correspondence explains the homological dimension bound and gives the local-system control needed to identify the universal-cover H_2 module.

The explicit families show that the distinction is not caused merely by changing the matrix or the kernel. The same scalar system can change from $S^1 \vee S^1$ to T^2 when only the residual changes, while another fixed-rank family has

$$\pi_1 \cong \mathbb{Z}^2$$

throughout but unbounded second Betti number. Thus the higher-rank completion topology is governed by optimal-fiber translation geometry rather than by rank alone.

Statements and Declarations

Author Contributions. Alp Eren Bütün: Conceptualization, methodology, formal analysis, investigation, software, validation, visualization, writing—original draft, and writing—review and editing.

Funding. No funding was received for conducting this study.

Competing Interests. The author has no relevant financial or non-financial interests to disclose.

Data and Code Availability. The paper is theoretical and uses no empirical dataset. All definitions, constructions, formulas, and proofs needed for the mathematical results are contained in the manuscript. No external dataset or executable code is required to establish the theoretical results. Computational consistency checks summarized in [section 13](#) are auxiliary validation tools and do not replace the symbolic proofs.

Ethics Approval and Consent. Not applicable.

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